

Omni-Depth: Monocular Metric Distance Estimation for Drone Targets Using Synthetic Data, Relative Depth, and Geometric Cues

Students: Noam Solow, Maoz Daniel

Academic Advisor: Dr. Aviram Zilberman

Company Worked With: Gurdian Angle

Company Advisor: Lior Shilon

Background

The growing use of drones has increased the need for systems that can detect them and determine their distance from an observing camera. Distance information is important for tracking, collision avoidance, surveillance, and counter-UAV applications. Existing solutions often use LiDAR, radar, or stereo cameras, but these add cost, weight, power consumption, and calibration requirements.

A monocular RGB camera is a standard imaging device that captures red, green, and blue color information from a single viewpoint. Unlike stereo vision systems, which estimate depth by computing disparity between two views, a monocular system provides only one perspective. As a result, although a single image contains visual cues such as relative object size, perspective, and scene structure, it inherently lacks explicit physical scale. Consequently, monocular depth estimation models typically produce a relative depth map, where each pixel is assigned a value indicating its depth relative to other pixels in the image. These values identify which regions are closer or farther from the camera but do not directly correspond to real-world metric distances. Therefore, additional processing or calibration is required to convert relative depth estimates into accurate metric distance measurements in meters.

Motivation

The project explores whether these visual clues can be converted into a useful metric distance using only a single RGB image. Such a solution could reduce the dependence on dedicated depth hardware and information transmitted by the target drone. Because large drone datasets with reliable distance labels are scarce, we also created a controlled synthetic environment to develop and evaluate the proposed approach.

Research Objective

We developed ***Omni-Depth***, which combines relative depth information, the detected drone's size and position in the image, and available environmental context to estimate its distance from the camera. With known camera intrinsics, this estimate may later support 3D back-projection, although full 3D localization was not evaluated.

The research question is: **Is it possible to estimate the metric distance between a drone and an observing camera from a single image, without using additional hardware or components?**

For each image, the system receives the detected drone region, relative-depth information, and available contextual metadata. It then predicts the drone's distance in meters. During training, the model learns to minimize the average absolute difference between its predictions and the known ground-truth distances. The process uses no stereo camera, LiDAR, radar, GPS transmission, or cooperative signal from the target.

Tools and Methods

- **Dataset creation:** An automated Unreal Engine pipeline generated 15,064 images over 20-150 m, varying viewpoint, weather, and time. The pipeline automatically captured images while simultaneously recording the drone's true distance from the camera, environmental metadata, and object annotations
- **Feature extraction:** A pretrained monocular model produced relative-depth maps. Local depth cues were combined with normalized box geometry, position, and environmental metadata. Box shifts and scaling were summarized by median and standard deviation for robustness.
- **Implementation:** The Python pipeline used PyTorch, OpenCV, NumPy, scikit-learn, and XGBoost; cached feature tables supported reproducible comparisons.
- **Models:** We examined global scale-only calibration, depth-only linear regression, linear regression with geometry and metadata, improved stacked linear regression, Random Forest, perturbation-aggregated Random Forest, XGBoost with the same robust features, and a weighted Random Forest-XGBoost ensemble (0.17/0.83).
- **Synthetic evaluation:** The stratified split contained 12,804 development and 2,260 held-out test samples. Five-fold cross-validation preceded final testing. Evaluation included *MAE*, *RMSE*, (R^2), error-tolerance rates, and breakdowns by distance and environment.
- **Real evaluation:** The selected model was tested on 489 real-world drone images collected by Nenrus. Global and per-drone-scale affine and quadratic calibration models were evaluated using more than 100 repeated stratified train-test splits with calibration fractions of 10%, 20%, 30%, and 50%.

Results

One of the key findings was that relative depth alone was insufficient for accurate metric distance estimation. Bounding-box geometry provided the strongest cue, while depth information further improved performance when combined with geometric features.

The final ensemble achieved **6.008 MAE**, ($R^2 = 0.949$), with **81.0%** of predictions within 10 meters on the synthetic benchmark.

During field experiments, initial predictions exposed systematic domain bias. Per-drone quadratic calibration fitted on 20% of the data and evaluated on the remainder achieved **3.278 MAE**, with **96.44%** of predictions within 10 meters, showing the value of limited target-domain data.

Conclusions and Contributions

Omni-Depth shows that relative depth and geometry can recover metric distance in the evaluated synthetic domain. The project contributes an automatically annotated dataset, a hybrid monocular framework, and systematic model evaluation. Direct synthetic-to-real transfer remained limited, but lightweight calibration corrected much of the bias without retraining.

A research paper presenting the Omni-Depth framework, dataset, methodology, and findings has been written and is expected to be published soon.